

A single kinetic defect in a run-and-tumble ring: exact spectrum, exceptional points, and relaxation crossover

Abstract. We solve the spectral problem for one continuous-time run-and-tumble particle on an n -site ring when the tumbling rate at one site is ω_0 and the bulk rate is ω . The defect changes the $2n \times 2n$ generator by rank one. The characteristic polynomial is the homogeneous polynomial plus $(\omega_0 - \omega)/n$ times its ω -derivative. This identity gives Chebyshev and transfer-matrix quantization formulas, a parity factorization, and finite polynomial tests for every Jordan block. We separately resolve the singular flat-band point $\omega = 1$, where the eigenvalue -2 can carry Jordan blocks of sizes two or three. The infinite-line Green function gives an exact criterion for exponentially localized defect modes. For fixed positive bulk rate, reflection protects one homogeneous long-wave mode; consequently a slow defect, including $\omega_0 \asymp n^{-1}$, does not produce a new spectral-gap scaling. The nontrivial n^{-1} crossover occurs when the bulk rate itself is of order n^{-1} . Finally, the standard telegraph equation misses the lattice Poisson-diffusion term at fixed ω , while a diffusion-corrected telegraph equation agrees through order n^{-2} .

Keywords. Run-and-tumble particle; kinetic defect; non-Hermitian spectrum; exceptional point; localized mode; spectral gap; telegraph equation.

1. Model and notation

Definition 1.1 (Defective run-and-tumble generator). Let $n \geq 2$, let $\omega > 0$, and let $\omega_0 \geq 0$. Sites are $j \in \mathbb{Z}/n\mathbb{Z}$, orientations are $\sigma \in \{+, -\}$, and

$$\omega_j = \begin{cases} \omega_0, & j = 0, \\ \omega, & j \neq 0. \end{cases}$$

The forward generator $L = L_n(\omega, \omega_0)$ is

$$(Lu)_j^+ = u_{j-1}^+ - (1 + \omega_j)u_j^+ + \omega_j u_j^-, \quad (1.1)$$

$$(Lu)_j^- = u_{j+1}^- - (1 + \omega_j)u_j^- + \omega_j u_j^+. \quad (1.2)$$

Thus hopping in the current orientation has rate one and tumbling has rate ω_j . Put

$$\delta = \omega_0 - \omega, \quad q_k = \frac{2\pi k}{n}, \quad c_k = \cos q_k, \quad s_k = \sin q_k.$$

Equations (1.1)–(1.2) use the same one-particle convention as equations (6)–(10) of [MBE19]. That paper records the homogeneous eigenvalues in its equation (11), the homogeneous exceptional points in equation (14), and the first-mode relaxation rate in equations (20)–(21).

Lemma 1.2 (*Rank-one defect*). *Let $L^{(0)} = L_n(\omega, \omega)$. If $d = e_{0,+} - e_{0,-}$, then*

$$L = L^{(0)} - \delta dd^\top.$$

In particular, the perturbation has rank one when $\delta \neq 0$.

Proof. At site zero, changing ω to ω_0 adds

$$\delta \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} = -\delta \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \end{pmatrix}.$$

Every other matrix entry is unchanged. ■

2. Three exact characteristic equations

Definition 2.1 (Homogeneous factors). For $\lambda \in \mathbb{C}$, set

$$a = \lambda + 1 + \omega, \tag{2.1}$$

$$x = \frac{a^2 + 1 - \omega^2}{2a}, \tag{2.2}$$

$$D_k(\lambda) = (\lambda + 1 - c_k)(\lambda + 1 + 2\omega - c_k) + s_k^2 = a^2 + 1 - \omega^2 - 2ac_k. \tag{2.3}$$

Although x is undefined at $a = 0$, each expression below containing $a^n T_n(x)$ is understood by its unique polynomial continuation.

Lemma 2.2 (*Homogeneous polynomial*). *The homogeneous characteristic polynomial is*

$$\begin{aligned} P_n^{(0)}(\lambda; \omega) &= \det(\lambda I - L^{(0)}) \\ &= \prod_{k=0}^{n-1} D_k(\lambda) \end{aligned} \tag{2.4}$$

$$= 2a^n (T_n(x) - 1), \tag{2.5}$$

where T_n is the Chebyshev polynomial of the first kind. The two roots of D_k are

$$\lambda_k^\pm = c_k - 1 - \omega \pm \sqrt{\omega^2 - s_k^2}.$$

Proof. Fourier transformation gives the block

$$W_k = \begin{pmatrix} e^{iq_k} - 1 - \omega & \omega \\ \omega & e^{-iq_k} - 1 - \omega \end{pmatrix}.$$

Its characteristic polynomial is D_k , which proves (2.4) and the displayed roots. Since $D_k = 2a(x - c_k)$ and

$$\prod_{k=0}^{n-1} (x - c_k) = 2^{1-n} (T_n(x) - 1),$$

equation (2.5) follows. ■

Theorem 2.3 (*Exact finite characteristic polynomial*). For every $n \geq 2$, $\omega > 0$, and $\omega_0 \geq 0$,

$$P_n(\lambda; \omega, \omega_0) := \det(\lambda I - L_n(\omega, \omega_0)) \quad (2.6)$$

$$= P_n^{(0)}(\lambda; \omega) + \frac{\delta}{n} \partial_\omega P_n^{(0)}(\lambda; \omega) \quad (2.7)$$

$$= P_n^{(0)}(\lambda; \omega) \left[1 + \frac{2\delta}{n} \sum_{k=0}^{n-1} \frac{\lambda + 1 - c_k}{D_k(\lambda)} \right]. \quad (2.8)$$

Equation (2.8) is interpreted by polynomial continuation at a homogeneous eigenvalue. Equivalently,

$$P_n = 2a^n(T_n(x) - 1) + 2\delta a^{n-1}(T_n(x) - 1) + \delta a^{n-2}\lambda(\lambda + 2)U_{n-1}(x), \quad (2.9)$$

where U_{n-1} is the Chebyshev polynomial of the second kind and the right-hand side again denotes its polynomial continuation.

Proof. Let $B = \lambda I - L^{(0)}$. Lemma 1.2 and the matrix determinant lemma give

$$\det(\lambda I - L) = \det B(1 + \delta d^\top B^{-1}d).$$

In Fourier space,

$$d^\top B^{-1}d = \frac{1}{n} \sum_{k=0}^{n-1} \frac{2(\lambda + 1 - c_k)}{D_k(\lambda)},$$

which proves (2.8). Direct differentiation of (2.4) gives

$$\partial_\omega P_n^{(0)} = P_n^{(0)} \sum_k \frac{2(\lambda + 1 - c_k)}{D_k},$$

and hence (2.7). Finally,

$$\partial_\omega x = \frac{\lambda(\lambda + 2)}{2a^2}, \quad T'_n(x) = nU_{n-1}(x),$$

and differentiating (2.5) proves (2.9). ■

Theorem 2.4 (*Transfer-matrix quantization*). For a local rate $r \geq 0$, define

$$T_r(\lambda) = \frac{1}{\lambda + 1 + r} \begin{pmatrix} 1 & r \\ -r & (\lambda + 1 + r)^2 - r^2 \end{pmatrix}.$$

Then $\det T_r = 1$, and, away from the displayed denominators,

$$P_n(\lambda; \omega, \omega_0) = a^{n-1}(\lambda + 1 + \omega_0) \{ \operatorname{tr}[T_\omega(\lambda)^{n-1} T_{\omega_0}(\lambda)] - 2 \}.$$

After multiplication by the denominators, this identity holds for every λ by polynomial continuation. Thus an eigenvalue is equivalently characterized by

$$\operatorname{tr}[T_\omega(\lambda)^{n-1} T_{\omega_0}(\lambda)] = 2.$$

Moreover, if $t = \text{tr } T_\omega = 2x$, then

$$T_\omega^{n-1} = U_{n-2}(x)T_\omega - U_{n-3}(x)I,$$

with $U_{-1} = 0$.

Proof. The eigenvalue equations at site j are

$$\begin{aligned} (\lambda + 1 + \omega_j)u_j^+ - \omega_j u_j^- &= u_{j-1}^+, \\ (\lambda + 1 + \omega_j)u_j^- - \omega_j u_j^+ &= u_{j+1}^-. \end{aligned}$$

Consequently

$$\begin{pmatrix} u_j^+ \\ u_{j+1}^- \end{pmatrix} = T_{\omega_j}(\lambda) \begin{pmatrix} u_{j-1}^+ \\ u_j^- \end{pmatrix}.$$

Periodic boundary conditions require the product monodromy to have eigenvalue one. Its determinant is one, so this is equivalent to trace two. Both sides of (2.4) are monic polynomials of degree $2n$ and have the same zeros with multiplicity; comparison as $\lambda \rightarrow \infty$ fixes the factor shown. Equation (2.4) is the Cayley–Hamilton identity for a unimodular 2×2 matrix. $\blacksquare$

3. Parity factorization and every finite exceptional point

Definition 3.1 (Kinetic reflection). Define the involution

$$(Ru)_j^+ = u_{-j}^-, \quad (Ru)_j^- = u_{-j}^+.$$

Write $\mathcal{H}_\pm = \ker(R \mp I)$.

Lemma 3.2 (Protected sector). *The generator commutes with R , the defect vector satisfies $Rd = -d$, and*

$$L|_{\mathcal{H}_+} = L^{(0)}|_{\mathcal{H}_+}.$$

Thus the defect changes only the reflection-odd sector.

Proof. Reflection reverses position and orientation simultaneously, leaving the directed hopping rules and the rate field ω_j invariant. At the fixed site zero it exchanges the two entries of d , hence $Rd = -d$. Lemma 1.2 then vanishes on $\mathcal{H}_+$. $\blacksquare$

Definition 3.3 (Parity polynomials). Put $m = \lfloor (n-1)/2 \rfloor$ and let ε_n be one for even n and zero for odd n . Define

$$H_{n,+}(\lambda) = \lambda(\lambda + 2)^{\varepsilon_n} \prod_{k=1}^m D_k(\lambda), \tag{3.1}$$

$$H_{n,-}^{(0)}(\lambda) = (\lambda + 2\omega)(\lambda + 2 + 2\omega)^{\varepsilon_n} \prod_{k=1}^m D_k(\lambda), \tag{3.2}$$

$$G_n(\lambda) = \frac{1}{n} \left[\frac{1}{\lambda + 2\omega} + \frac{\varepsilon_n}{\lambda + 2 + 2\omega} + 2 \sum_{k=1}^m \frac{\lambda + 1 - c_k}{D_k(\lambda)} \right], \tag{3.3}$$

$$Q_{n,-}(\lambda) = H_{n,-}^{(0)}(\lambda) (1 + 2\delta G_n(\lambda)). \tag{3.4}$$

Every quotient in (3.4) cancels, so $Q_{n,-}$ is a monic polynomial of degree n . An entirely polynomial version is

$$Q_{n,-} = H_{n,-}^{(0)} + \frac{2\delta}{n} \left[\frac{H_{n,-}^{(0)}}{\lambda + 2\omega} + \varepsilon_n \frac{H_{n,-}^{(0)}}{\lambda + 2 + 2\omega} + 2 \sum_{k=1}^m (\lambda + 1 - c_k) \frac{H_{n,-}^{(0)}}{D_k} \right]. \quad (3.5)$$

Theorem 3.4 (*Exact parity factorization*). *The characteristic polynomial factors as*

$$P_n(\lambda; \omega, \omega_0) = H_{n,+}(\lambda) Q_{n,-}(\lambda).$$

The first factor is the characteristic polynomial on $\mathcal{H}_+$; the second is the characteristic polynomial on $\mathcal{H}_-$.

Proof. The modes q_k and $-q_k$ form a four-dimensional Fourier subspace, which splits into two isomorphic two-dimensional R -eigenspaces. Each has characteristic polynomial D_k . At $q = 0$, the vectors $(1, 1)$ and $(1, -1)$ have parities $+$ and $-$ and eigenvalues 0 and -2ω . When n is even, the $q = \pi$ density and polarization vectors have parities $+$ and $-$ and eigenvalues -2 and $-2 - 2\omega$. This gives (3.1)–(3.2). Lemma 3.2 and the rank-one determinant lemma restricted to $\mathcal{H}_-$ give (3.4). Multiplication proves (3.4). ■

Definition 3.5 (Exceptional point). For an eigenvalue μ , let $a(\mu)$ be its algebraic multiplicity and $g(\mu) = \dim \ker(\mu I - L)$ its geometric multiplicity. We call (ω, ω_0, μ) an exceptional point when $a(\mu) > g(\mu)$. Write $J_r(\mu)$ for a Jordan block of size r at μ .

Lemma 3.6 (*No inter-wave coincidences away from the flat band*). *If $c_k \neq c_\ell$ and $D_k(\mu) = D_\ell(\mu) = 0$, then $\omega = 1$ and $\mu = -2$.*

Proof. Subtracting the two equations gives $2a(c_k - c_\ell) = 0$, hence $a = 0$. Substitution into (2.3) gives $1 - \omega^2 = 0$. Since $\omega > 0$, this means $\omega = 1$ and then $\mu = -1 - \omega = -2$. ■

Theorem 3.7 (*Jordan classification for $\omega \neq 1$*). *Assume $\omega \neq 1$.*

1. *In the protected sector, the endpoint factors in (3.1) give one-dimensional blocks. For each $1 \leq k \leq m$, a root of D_k gives one J_1 , except that*

$$\omega = |s_k|, \quad \mu = c_k - 1 - \omega. \quad (3.6)$$

gives one $J_2(\mu)$.

2. *In the affected sector, if*

$$Q_{n,-}(\mu) = Q'_{n,-}(\mu) = \cdots = Q_{n,-}^{(r-1)}(\mu) = 0, \quad Q_{n,-}^{(r)}(\mu) \neq 0, \quad (3.7)$$

then the complete affected-sector contribution at μ is one block $J_r(\mu)$.

3. The full Jordan form at μ is the direct sum of the blocks in items 1 and 2. In particular, a crossing between the two parity sectors is semisimple unless one of its sector blocks has size greater than one.

Consequently, (3.6) together with (3.7) is an exact finite criterion for every exceptional point when $\omega \neq 1$. The order r in (3.7) classifies exceptional points of arbitrary order.

Proof. The discriminant of D_k is $4(\omega^2 - s_k^2)$. At a zero discriminant its Fourier block is not scalar and hence is a single J_2 ; otherwise each root is simple. Lemma 3.6 excludes coincidences between distinct c_k values.

It remains to show that the affected restriction is cyclic. In each two-dimensional homogeneous parity block, the spectral weight of the defect vector has numerator $2(\lambda + 1 - c_k)$. This numerator and D_k have no common root for an interior wave number: a common root would give $\lambda + 1 = c_k$, after which $D_k = s_k^2 \neq 0$. The endpoint weights in (3.3) are also nonzero. Lemma 3.6 makes the block minimal polynomials coprime. The defect vector is therefore cyclic for the homogeneous odd restriction. If A has cyclic vector d , then d is also cyclic for $A - \delta dd^\top$, because each $(A - \delta dd^\top)^j d$ equals $A^j d$ plus a linear combination of lower powers. Hence the minimal and characteristic polynomials of the affected restriction both equal $Q_{n,-}$. A cyclic matrix has one Jordan block for each eigenvalue, of size equal to the root order. The direct-sum assertion follows from the invariant parity splitting. ■

Theorem 3.8 (*The complete flat-band classification at $\omega = 1$*). Let $\omega = 1$ and put $\mu = -2$.

1. If $\omega_0 = 1$ and $4 \nmid n$, the contribution at μ is $nJ_1(\mu)$.
2. If $\omega_0 = 1$ and $4 \mid n$, the contribution is

$$2J_2(\mu) \oplus (n-2)J_1(\mu).$$

3. If $\omega_0 \neq 1$ and $n \equiv 1$ or $2 \pmod{4}$, the contribution is $(n-1)J_1(\mu)$.
4. If $\omega_0 \neq 1$ and $n \equiv 0 \pmod{4}$, the contribution is

$$J_2(\mu) \oplus (n-2)J_1(\mu).$$

5. If $n \equiv 3 \pmod{4}$, $\omega_0 \neq 1$, and $\omega_0 \neq (n-1)/(n+1)$, the contribution is again $J_2(\mu) \oplus (n-2)J_1(\mu)$.
6. If $n \equiv 3 \pmod{4}$ and $\omega_0 = (n-1)/(n+1)$, the contribution is

$$J_3(\mu) \oplus (n-2)J_1(\mu).$$

For eigenvalues $\lambda \neq -2$, every protected-sector root is a one-dimensional block. Remove the exact power of $(\lambda + 2)$ specified above from $Q_{n,-}$. The remaining affected polynomial is cyclic, so the derivative test (3.7) applied to that reduced polynomial classifies every remaining affected block at $\omega = 1$. The full Jordan form is the direct sum of the protected and affected blocks.

Proof. Write $t = \lambda + 2$. At $\omega = 1$,

$$D_k = t(t - 2c_k), \quad P_n = t^{n-1}F_n(t),$$

where, for $\delta = \omega_0 - 1$,

$$F_n(t) = 2t(T_n(t/2) - 1) + \delta [2(T_n(t/2) - 1) + (t - 2)U_{n-1}(t/2)]. \quad (3.8)$$

The values

$$T_n(0) = \cos(n\pi/2), \quad U_{n-1}(0) = \sin(n\pi/2)$$

give the following algebraic multiplicities. For $\delta = 0$ they are n , except that $4 \mid n$ gives $n + 2$. For $\delta \neq 0$ they are $n - 1$ when $n \equiv 1, 2 \pmod{4}$, and n when $n \equiv 0, 3 \pmod{4}$. In the last congruence class, $F'_n(0) = -2 - \delta(n + 1)$; it vanishes exactly at $\delta = -2/(n + 1)$, when $F''_n(0) \neq 0$ and the multiplicity is $n + 1$.

The bulk equations at $t = 0$ read

$$u_j^- = -u_{j-1}^+ \quad (j \neq 0).$$

Together they impose this relation for every j . The defect equation adds no constraint when $\omega_0 = 1$ and otherwise adds the single constraint $u_0^+ + u_{n-1}^+ = 0$. Thus the geometric multiplicity is n in the homogeneous case and $n - 1$ otherwise. Fourier decomposition shows that the two excess algebraic dimensions when $4 \mid n$ and $\delta = 0$ are the two $q = \pm\pi/2$ blocks, each a J_2 . For $\delta \neq 0$, the parity factorization places the $4 \mid n$ excess in one protected J_2 . When $n \equiv 3 \pmod{4}$, the affected block is one J_2 generically; the additional order at $\delta = -2/(n + 1)$ extends it to J_3 . These counts give all six cases. After the flat factor is removed, the coprime-block cyclicity argument from Theorem 3.7 applies unchanged. ■

Corollary 3.9 (*Finite algebraic exceptional set*). *For fixed n , all non-flat affected exceptional parameters lie on the explicit algebraic set*

$$\text{Res}_\lambda(Q_{n,-}, \partial_\lambda Q_{n,-}) = 0,$$

with $Q_{n,-}$ given by (3.5). At $\omega = 1$, the flat factor is removed before taking the resultant. Root orders and theorems 3.7–3.8 distinguish exceptional points from semisimple parity crossings without a numerical tolerance.

Proof. The resultant vanishes exactly when the two displayed polynomials share a root. The preceding two theorems identify the Jordan block attached to every such root and separately account for the flat factor. ■

4. Defect-localized modes

Definition 4.1 (Bulk spectral set). For the infinite translation-invariant chain, define

$$\Sigma_\omega = \left\{ c - 1 - \omega \pm \sqrt{\omega^2 - (1 - c^2)} : c \in [-1, 1] \right\}.$$

For $\lambda \notin \Sigma_\omega$, retain a and x from (2.1)–(2.2), and choose

$$s(\lambda) = \sqrt{x(\lambda)^2 - 1}$$

as the branch satisfying $s(\lambda) \sim x(\lambda)$ as $|\lambda| \rightarrow \infty$. Put $z = x - s$, so $|z| < 1$.

Theorem 4.2 (*Infinite-line localization criterion*). *Let $\delta \neq 0$. A number $\lambda \notin \Sigma_\omega$ is an exponentially localized eigenvalue of the infinite chain with the same single tumbling defect if and only if*

$$1 + 2\delta g_\infty(\lambda) = 0, \quad (4.1)$$

where

$$g_\infty(\lambda) = \frac{1}{2a} \left(1 + \frac{\lambda + 1 - x}{s} \right) = \frac{z(\lambda + 1 - z)}{a(1 - z^2)}. \quad (4.2)$$

The localization length is

$$\xi(\lambda) = \frac{-1}{\log |z|}.$$

Equivalently, with $a = \lambda + 1 + \omega$, the pair (a, z) satisfies

$$a^2 + 1 - \omega^2 - a(z + z^{-1}) = 0, \quad (4.3)$$

$$a(1 - z^2) + 2\delta z(a - \omega - z) = 0, \quad 0 < |z| < 1. \quad (4.4)$$

After elimination of a , every admissible z is a root of

$$\begin{aligned} 0 &= 2\delta z^3 + (2\delta\omega + 1 - \omega^2)z^2 \\ &\quad + (4\delta^2\omega + 4\delta\omega^2 - 2\delta)z + (2\delta\omega + \omega^2 - 1), \end{aligned} \quad (4.5)$$

and, when the denominator is nonzero,

$$a = \frac{2\delta z(\omega + z)}{1 - z^2 + 2\delta z}, \quad \lambda = a - 1 - \omega. \quad (4.6)$$

A root of (4.5) is accepted only when (4.1), or equivalently both (4.3) and (4.4), is satisfied; this removes roots introduced by elimination or by clearing denominators.

Proof. The polarization entry of the bulk resolvent at the defect site is

$$\begin{aligned} g_\infty(\lambda) &= \frac{1}{2\pi} \int_0^{2\pi} \frac{\lambda + 1 - \cos q}{a^2 + 1 - \omega^2 - 2a \cos q} dq \\ &= \frac{1}{2a} \left[1 + (\lambda + 1 - x) \frac{1}{2\pi} \int_0^{2\pi} \frac{dq}{x - \cos q} \right] \\ &= \frac{1}{2a} \left(1 + \frac{\lambda + 1 - x}{s} \right). \end{aligned}$$

The rank-one eigenvalue equation is (4.1). Since $x = (z + z^{-1})/2$ and $s = (z^{-1} - z)/2$, elementary simplification gives the second expression in (4.2), and the Green kernel decays as $z^{|j|}$. Clearing the denominator in (4.1) gives (4.4); the bulk dispersion relation gives (4.3). Eliminating a gives (4.5), while solving (4.4) for a gives (4.6). ■

Proposition 4.3 (*Finite-ring convergence of a localized mode*). Suppose (4.1) has a simple root λ_∞ with decay root z_∞ , $|z_\infty| < 1$. Then, for all sufficiently large n , the finite polynomial $Q_{n,-}$ has a unique root λ_n near λ_∞ , and

$$\lambda_n - \lambda_\infty = O(|z_\infty|^n).$$

The corresponding right eigenvector has site mass bounded by $C|z_\infty|^{2 \operatorname{dist}(j,0)}$ after normalization.

Proof. The finite Green function (3.3) is the periodization of the infinite Green kernel. On a compact neighborhood of a point outside Σ_ω , its image terms are bounded by a geometric series with first term $|z_\infty|^n$. Hence $G_n - g_\infty = O(|z_\infty|^n)$ uniformly there. The simple-root implicit-function estimate applied to $1 + 2\delta G_n$ gives the eigenvalue bound. The resolvent representation of the eigenvector and the same image estimate give the spatial bound. ■

5. Large-ring spectral gap

Definition 5.1 (Spectral gap). For $\omega > 0$ and $n \geq 2$, irreducibility makes zero a simple eigenvalue. Define

$$\gamma_n(\omega, \omega_0) = -\max\{\Re \lambda : \lambda \in \operatorname{spec} L_n(\omega, \omega_0), \lambda \neq 0\}.$$

Put $q = 2\pi/n$ and

$$\gamma_n^{\operatorname{hom}}(\omega) = 1 - \cos q + \omega - \sqrt{\omega^2 - \sin^2 q}, \quad (5.1)$$

where the square root is positive once n is sufficiently large for fixed $\omega > 0$.

Lemma 5.2 (*Protected first mode and affected pole*). For every n , the protected sector contains $\lambda_1^+ = -\gamma_n^{\operatorname{hom}}$. For fixed positive ω and ω_0 , the affected root born from the same homogeneous pole is

$$\lambda_{1,-} = \lambda_1^+ + \frac{\delta(1+\omega)}{\omega^2(1+\omega_0)} \frac{q^2}{n} + O(n^{-4}). \quad (5.2)$$

Proof. The first assertion is Theorem 3.4. Near λ_1^+ , the two terms $k = 1, n-1$ in the full Fourier Green function have residue

$$R_1 = \frac{\lambda_1^+ + 1 - c_1}{\lambda_1^+ - \lambda_1^-} = -\frac{\omega - r_1}{2r_1}, \quad r_1 = \sqrt{\omega^2 - \sin^2 q}.$$

Thus

$$G_n(\lambda) = \frac{2R_1}{n(\lambda - \lambda_1^+)} + G_n^{\operatorname{reg}}(\lambda).$$

The regular part tends to $1/[2(1+\omega)]$. Solving $1 + 2\delta G_n = 0$ gives

$$\lambda_{1,-} - \lambda_1^+ = \frac{2\delta(\omega - r_1)}{nr_1} \frac{1}{1 + \delta/(1+\omega)} + O(n^{-4}),$$

which expands to (5.2). ■

Theorem 5.3 (*Fixed-rate gap asymptotics*). Fix $\omega > 0$ and $\omega_0 \geq 0$.

1. If $\omega_0 \leq \omega$, then for every sufficiently large n ,

$$\gamma_n(\omega, \omega_0) = \gamma_n^{\text{hom}}(\omega). \quad (5.3)$$

2. If $\omega_0 > \omega$, then

$$\gamma_n(\omega, \omega_0) = \gamma_n^{\text{hom}}(\omega) - \frac{4\pi^2(\omega_0 - \omega)(1 + \omega)}{\omega^2(1 + \omega_0)} n^{-3} + O(n^{-4}). \quad (5.4)$$

3. In both cases,

$$\begin{aligned} \gamma_n^{\text{hom}}(\omega) &= \frac{2\pi^2(1 + \omega)}{\omega} n^{-2} \\ &\quad - 16\pi^4 \left(\frac{1}{24} + \frac{1}{6\omega} - \frac{1}{8\omega^3} \right) n^{-4} + O(n^{-6}). \end{aligned} \quad (5.5)$$

The errors are uniform when (ω, ω_0) ranges over a compact set with ω bounded away from zero; in (5.3) the ordering is uniform away from $\omega_0 = \omega$.

Proof. Taylor expansion of (5.1) gives (5.5). Lemma 5.2 shows that the affected first root moves toward zero precisely when $\delta > 0$. The protected and affected roots associated with wave number two have real parts separated from the first pair by a positive multiple of n^{-2} ; all remaining extended roots are farther left, and Proposition 4.3 places every fixed-rate localized root a positive distance from zero. These estimates are uniform on the stated compact sets. Therefore the protected root wins when $\delta \leq 0$, proving (5.3), and the affected root wins when $\delta > 0$. Substitution of $q^2 = 4\pi^2 n^{-2}$ in (5.2) then gives (5.4). ■

Corollary 5.4 (*Fixed ratio, vanishing ratio, and a slow-site scaling*). Let the bulk rate $\omega > 0$ be fixed.

1. If $\omega_0/\omega \rightarrow \rho \in (0, \infty)$, equations (5.3)–(5.5) apply. For $\rho < 1$ the protected branch is selected; for $\rho > 1$ the affected branch and its n^{-3} term are selected. The boundary $\rho = 1$ is decided by the eventual sign and size of $\omega_0 - \omega$.
2. If $\omega_0/\omega \rightarrow 0$, then (5.3) holds for every sufficiently large n .
3. In particular, if $\omega_0 \sim \beta/n$ with $\beta \geq 0$, then (5.3) still holds. A single slow tumbling site therefore has no separate $\omega_0 \asymp n^{-1}$ gap crossover when the bulk rate is fixed.

Proof. In items 2 and 3 one has $\omega_0 < \omega$ for every sufficiently large n , so Theorem 5.3 applies. Item 1 follows by the same sign test and compact-uniform expansion. ■

Theorem 5.5 (*Bulk-critical double scaling*). Set

$$\omega = \frac{\alpha}{n}, \quad \omega_0 = \frac{\beta}{n}, \quad k = 2\pi,$$

where $\alpha > 0$ and $\beta \geq 0$ are fixed. Away from $\alpha = k$,

$$\alpha < k : \quad \gamma_n = \frac{\alpha}{n} + \left[\frac{k^2}{2} - 2(\alpha - \beta)_+ \right] n^{-2} + O(n^{-3}), \quad (5.6)$$

$$\begin{aligned} \alpha > k : \quad \gamma_n &= \frac{\alpha - \sqrt{\alpha^2 - k^2}}{n} \\ &+ \left[\frac{k^2}{2} - 2(\beta - \alpha)_+ + \frac{\alpha - \sqrt{\alpha^2 - k^2}}{\sqrt{\alpha^2 - k^2}} \right] n^{-2} + O(n^{-3}). \end{aligned} \quad (5.7)$$

At $\alpha = k$,

$$\beta \leq k : \quad \gamma_n = \frac{k}{n} + k^2 \left(\frac{1}{2} - \frac{1}{\sqrt{3}} \right) n^{-2} + O(n^{-3}), \quad (5.8)$$

$$\beta > k : \quad \gamma_n = \frac{k}{n} - 2\sqrt{k(\beta - k)} n^{-3/2} + O(n^{-2}). \quad (5.9)$$

Thus the first homogeneous exceptional scaling is a square-root boundary layer for a faster defect, while the side selecting the affected parity mode reverses as α passes through 2π .

Proof. For $\alpha < k$, the protected first pair has

$$\Re \lambda_1^+ = -\frac{\alpha}{n} - \frac{k^2}{2n^2} + O(n^{-3}).$$

The pole residue tends to $R_1 = \frac{1}{2} + i\alpha/(2\sqrt{k^2 - \alpha^2})$, and $\delta = (\beta - \alpha)/n$. Hence the affected real part shifts by $-2(\beta - \alpha)n^{-2} + O(n^{-3})$. Taking the larger real part gives (5.6).

For $\alpha > k$, put $s = \sqrt{\alpha^2 - k^2}$. The protected rate is $(\alpha - s)n^{-1} + k^2(2n^2)^{-1} + O(n^{-3})$, while $R_1 \rightarrow -(\alpha - s)/(2s)$. The same pole equation shifts the affected eigenvalue by $2(\beta - \alpha)(\alpha - s)/(sn^2) + O(n^{-3})$. Selection of the larger real part gives (5.7). The first wave number remains separated at order n^{-1} from all higher wave numbers in this case; for $\alpha < k$ the order- n^{-2} term separates it from them.

At $\alpha = k$, the exact lattice splitting of the homogeneous first block is

$$\sqrt{\omega^2 - \sin^2(k/n)} = \frac{k^2}{\sqrt{3}n^2} + O(n^{-4}),$$

which proves (5.8) when the protected root wins. Near the center of this block, writing the eigenvalue displacement as Δ , the two $q = \pm k/n$ poles give

$$G_n(\lambda) = -\frac{2k}{n^2\Delta^2}(1 + o(1)).$$

For $\beta > k$, the equation $1 + 2\delta G_n = 0$ has the positive displacement

$$\Delta = 2\sqrt{k(\beta - k)}n^{-3/2} + O(n^{-2}),$$

which gives (5.9). For $\beta < k$ the displacement is imaginary at this order and the protected lattice-split root remains closer to zero, giving (5.8). $\blacksquare$

6. Telegraph comparison

Definition 6.1 (Two continuum approximations). For a homogeneous ring of circumference n , the standard telegraph system is

$$\partial_t \rho = -\partial_x m, \quad \partial_t m = -\partial_x \rho - 2\omega m. \quad (6.1)$$

Its slow Fourier eigenvalue at wave number q is

$$\lambda_{\text{tel}}^+(q) = -\omega + \sqrt{\omega^2 - q^2}. \quad (6.2)$$

The lattice-diffusion-corrected system is

$$\partial_t \rho = -\partial_x m + \frac{1}{2} \partial_x^2 \rho, \quad \partial_t m = -\partial_x \rho + \frac{1}{2} \partial_x^2 m - 2\omega m. \quad (6.3)$$

whose slow eigenvalue is

$$\lambda_{\text{td}}^+(q) = -\frac{q^2}{2} - \omega + \sqrt{\omega^2 - q^2}. \quad (6.4)$$

Theorem 6.2 (*Exact relaxation rate versus telegraph rates*). Fix $\omega > 0$, let $q = 2\pi/n$, and define $\gamma_{\text{tel}} = -\lambda_{\text{tel}}^+(q)$ and $\gamma_{\text{td}} = -\lambda_{\text{td}}^+(q)$. Then

$$\gamma_n^{\text{hom}} - \gamma_{\text{tel}} = \frac{q^2}{2} - \left(\frac{1}{24} + \frac{1}{6\omega} \right) q^4 + O(q^6), \quad (6.5)$$

$$\gamma_n^{\text{hom}} - \gamma_{\text{td}} = -\left(\frac{1}{24} + \frac{1}{6\omega} \right) q^4 + O(q^6). \quad (6.6)$$

Consequently

$$\frac{\gamma_{\text{tel}}}{\gamma_n^{\text{hom}}} \rightarrow \frac{1}{1+\omega}, \quad \frac{\gamma_{\text{td}}}{\gamma_n^{\text{hom}}} \rightarrow 1. \quad (6.7)$$

In the bulk-critical scaling $\omega = \alpha/n$, the standard telegraph rate does recover the leading term:

$$\gamma_{\text{tel}} = \frac{\alpha - \Re \sqrt{\alpha^2 - (2\pi)^2}}{n}, \quad (6.8)$$

which agrees with the leading terms of (5.6)–(5.9).

Proof. The exact lattice dispersion replaces $-q^2/2$ by $\cos q - 1$ and q inside the square root by $\sin q$. Expanding $\cos q$, $\sin^2 q$, and the square root gives (6.5)–(6.6). Dividing by the leading exact rate $(1+\omega)q^2/(2\omega)$ gives (6.7). Substitution of $\omega = \alpha/n$ and $q = 2\pi/n$ into (6.2) gives (6.8). ■

Proposition 6.3 (*Point-defect telegraph secular equation*). Add $-2\delta \delta_0(x)m$ to the second equation in (6.1), with periodic boundary conditions on a circle of length n . Outside the homogeneous telegraph spectrum, put

$$\kappa = \sqrt{\lambda(\lambda + 2\omega)}$$

with positive real part. The affected-parity eigenvalues satisfy

$$1 + \delta \frac{\lambda}{\kappa} \coth \left(\frac{n\kappa}{2} \right) = 0. \quad (6.9)$$

The reflection-even telegraph modes are unchanged. Formula (6.9) is the continuum counterpart of (2.8); the exact lattice formula retains $1 - \cos q$ and $\sin q$, and (6.5)–(6.6) quantify the resulting relaxation-rate error.

Proof. The polarization entry of the telegraph resolvent is

$$\frac{1}{n} \sum_{\ell \in \mathbb{Z}} \frac{\lambda}{\lambda(\lambda + 2\omega) + (2\pi\ell/n)^2} = \frac{\lambda}{2\kappa} \coth\left(\frac{n\kappa}{2}\right).$$

Applying the rank-one determinant identity gives (6.9). The defect vector is reflection odd, so the even modes are protected exactly as in Lemma 3.2. ■

7. Conclusions

Proposition 7.1 (*The theorem package*). For the generator in Definition 1.1:

1. Theorem 2.3 and Theorem 2.4 give exact characteristic equations for every n , ω , and ω_0 .
2. Theorems 3.7 and 3.8 classify the algebraic and geometric multiplicities of every eigenvalue, including every exceptional point.
3. Theorem 4.2 gives an exact criterion and decay length for a defect-localized mode.
4. Theorems 5.3 and 5.5 give the fixed-rate, vanishing-defect, and bulk-critical large- n gap asymptotics.
5. Theorem 6.2 and Proposition 6.3 compare the exact relaxation rate with the standard and diffusion-corrected telegraph approximations.

Proof. Each item is the cited result. Their hypotheses all match Definition 1.1; no additional parameter is introduced. ■

AI-use disclosure

The proof architecture and manuscript were developed with assistance from OpenAI GPT-5.6 Sol (`gpt-5.6-sol`) at max reasoning effort. The model served as an interactive research assistant for deriving and testing arguments, checking citations and calculations, writing exact verification code, and editing the exposition. The author set the research direction, evaluated and approved the final arguments, and assumes responsibility for the mathematical claims, references, computations, and text.

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